

Sprint 1 Developer Test Case Table

This developer test case table documents the Sprint 1 automated TDD, unit, and API tests created and executed at code level. The tests were executed using Jest through npm test. The terminal evidence confirms 3 passed test suites and 24 passed automated tests.

Evidence reference for all rows: terminal screenshot showing npm test completed successfully with 3 passed test suites and 24 passed tests.

Test ID	Related User Story	Feature/Module	Test Scenario	Test Data	Test Steps / Test Script	Expected Result	Actual Result	Status	Evidence Reference
UT-01	User registration/login	Email validation	Reject invalid email addresses	student@gmail.com; student@icbt.lk	Run automated validator test in server/___tests___/validators.test.js	Only valid ICBT email is accepted	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-02	User registration	Password validation	Reject weak password and mismatched passwords	short; mismatched confirm password	Run automated validator test in server/___tests___/validators.test.js	Validation errors returned	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-03	User registration	Registration API	Block duplicate email registration	Existing registered email	Run API test in server/___tests___/sprint1-api.test.js	API returns duplicate email error	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
TDD-01	User registration	Registration API	Create account with valid details	Valid ICBT email, password, profile data	Run API test in server/___tests___/sprint1-api.test.js	Account, token, and profile are created	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
TDD-02	User registration	Registration API	Reject registration with invalid email	Invalid email address	Run API test in server/___tests___/sprint1-api.test.js	API returns validation error	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-04A	Authentication	Password security	Hash passwords and verify without storing plain text	Password123	Run auth unit test in server/___tests___/auth.test.js	Password hash differs from plain text and verifies correctly	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-04B	User registration	Password security	Store hashed registration password	New user registration payload	Run API test in server/___tests___/sprint1-api.test.js	Database stores hash, not plain password	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-05	User login	Login API	Authenticate valid login credentials	Existing email and correct password	Run API test in server/___tests___/sprint1-api.test.js	API returns 200 and token	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-06	User login	Login API	Reject incorrect password	Existing email and wrong password	Run API test in server/___tests___/sprint1-api.test.js	API returns unauthorized response	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-07	Authentication	Protected routes	Deny access without token	No authorization token	Run API test in server/___tests___/sprint1-api.test.js	API returns unauthorized response	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
TDD-06	Profile handling	Profile API	Display authenticated user's own profile	Valid driver token	Run API test in server/___tests___/sprint1-api.test.js	Own user/profile data is returned	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
TDD-07	Profile handling	Profile API	Update valid profile details	Valid profile update payload	Run API test in server/___tests___/sprint1-api.test.js	Profile and user name update successfully	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-08A	Profile handling	Profile validation	Reject invalid profile data	Empty first name / phone number	Run automated validator test in server/___tests___/validators.test.js	Validation errors returned	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-08B	Profile handling	Profile API	Reject invalid profile update data	Invalid profile payload	Run API test in server/___tests___/sprint1-api.test.js	API returns validation error	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-09	Privacy access control	Profile privacy	Deny access to another user's profile	Driver token requesting passenger profile	Run API test in server/___tests___/sprint1-api.test.js	API returns forbidden response	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-10A	Ride-offer creation	Ride validation	Reject ride offer with zero seats	availableSeats: 0	Run automated validator test in server/___tests___/validators.test.js	Validation error returned	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-10B	Ride-offer creation	Ride API	Reject ride offer with zero seats	Driver token, zero seats	Run API test in server/___tests___/sprint1-api.test.js	API returns validation error	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-11	Ride-offer creation	Ride API	Reject ride offer with negative seats	Driver token, availableSeats: -1	Run API test in server/___tests___/sprint1-api.test.js	API returns validation error	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-12	Ride-offer creation	Ride API	Reject ride offer missing origin	Empty origin	Run API test in server/___tests___/sprint1-api.test.js	API returns validation error	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-13	Ride-offer creation	Ride API	Reject ride offer missing destination	Empty destination	Run API test in server/___tests___/sprint1-api.test.js	API returns validation error	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-14	Ride-offer creation	Ride API	Link created ride offer to authenticated driver	Valid ride offer payload + driver token	Run API test in server/___tests___/sprint1-api.test.js	Created offer contains authenticated driver ID	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-15	Ride-offer listing	Ride API	Return only active offers for authenticated driver	Driver token with active/completed offers	Run API test in server/___tests___/sprint1-api.test.js	Only active own offers are returned	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
UT-16	Authorization	Driver-only access	Deny passenger access to ride-offer creation	Passenger token + ride payload	Run API test in server/___tests___/sprint1-api.test.js	API returns forbidden response	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed
API-17	Privacy access control	Ride details API	Return only authenticated driver's own ride detail	Own ride ID and another driver's ride ID	Run API test in server/___tests___/sprint1-api.test.js	Own offer returns success; other driver's offer is blocked	Passed in Jest npm test	Passed	Terminal screenshot: npm test passed

Automated Test Execution Evidence

Terminal screenshot captured after executing npm test. It shows all Sprint 1 developer test suites and tests passed successfully.

```
PS C:\Users\misal\OneDrive\Documents\git remote\ICBT_Carpooling_System> npm test

> icbt-carpooling-system@1.0.0 test
> node --experimental-vm-modules ./node_modules/jest/bin/jest.js --runInBand

(node:38888) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS server/__tests__/sprint1-api.test.js (37.696 s)
PASS server/__tests__/validators.test.js
PASS server/__tests__/auth.test.js

Test Suites: 3 passed, 3 total
Tests:       24 passed, 24 total
Snapshots:   0 total
Time:        41.033 s
Ran all test suites.
❖❖ PS C:\Users\misal\OneDrive\Documents\git remote\ICBT_Carpooling_System> 
```